MAVUNO HOME & CARE

# Revenue Decline Diagnostic

Why group revenue fell 10.7% — and who owns each cause

| | |
|---|---|
| Comparison period | Q3 2026 vs Q3 2025 |
| Prepared | 2026-10-05 |
| Prepared by | Senior Data Analyst |
| Classification | Internal — Commercial leadership |
| Reproduce with | python3 src/run_pipeline.py |

**The question this answers**

Whose plan gets rewritten next quarter — retention, pricing, or merchandising?

Group net revenue fell KSh 5.21M (10.7%). We did not lose customers — active customers rose +5.5%. We lost trips (-11.4% orders per customer) and basket (-4.5% average order value), and 71% of the damage sits in two of six stores.

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SECTION 1

# Executive summary

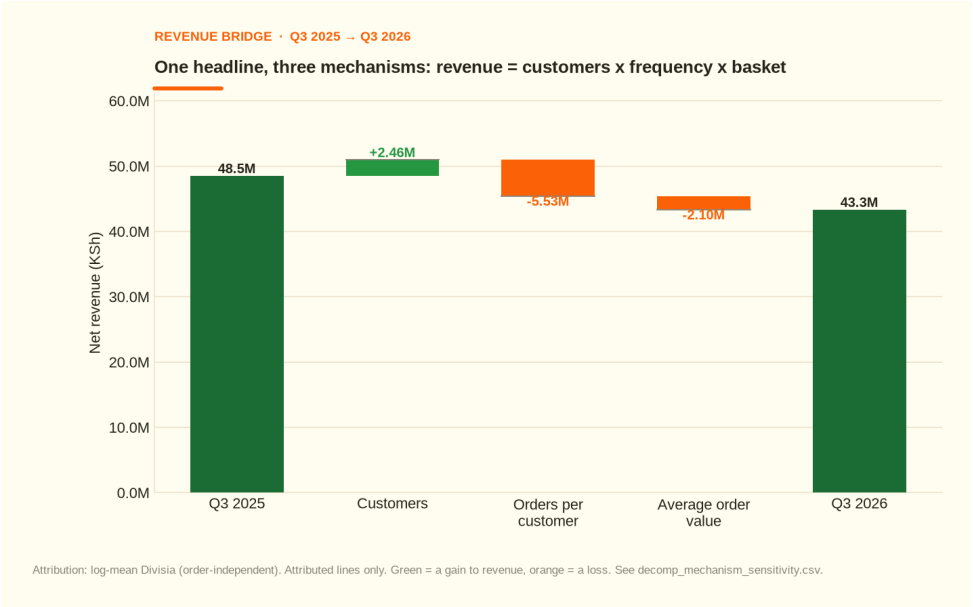
| NET REVENUE | GROSS MARGIN | ORDERS PER CUSTOMER |
|---|---|---|
| **KSh 43.65M** | **18.57%** | **4.82** |
| **-10.7%** from KSh 48.86M in Q3 2025 | **-1.38pp** gross profit KSh 8.04M | **-11.4%** the engine of the decline |
| **AVERAGE ORDER VALUE** | **ACTIVE CUSTOMERS** | **REVENUE WITH NO CUSTOMER ID** |
| **KSh 1,619** | **5,555** | **0.76%** |
| **-4.5%** 3.40 lines per order | **+5.5%** acquisition is NOT the problem | **KSh 0.70M** the declared coverage limit |

Group net revenue fell KSh 5.21M (10.7%). We did not lose customers — active customers rose +5.5%. We lost trips (-11.4% orders per customer) and basket (-4.5% average order value), and 71% of the damage sits in two of six stores.

Gross profit fell 16.9% against a 10.7% revenue fall, because gross margin compressed 1.38pp to 18.57%. Revenue is not profit: every recommendation below is therefore priced in gross profit, not revenue.

## The four findings, ranked by how much of the decline they own

| # | Owner | The problem | Cost | Share |
|---|---|---|---|---|
| 1 | Supply Chain & Merchandising | Two hero Personal Care SKUs out of stock for six weeks at Mombasa Nyali; the damage spread to whole baskets | KSh -2.16M | 41% |
| 2 | Retail Operations (Thika Road) | Thika Road footfall cut for the full quarter — visits per customer well below every other store | KSh -1.49M | 29% |
| 3 | CRM / Loyalty & Assortment | Residual: fewer trips per customer and slimmer baskets right across the chain | KSh -0.97M | 19% |
| 4 | Pricing & Category Management | Beverage promotional depth deepened sharply outside the two affected stores | KSh -0.55M | 11% |




*The 10.7% fall is one customer gain against two larger losses — it is not one event.*

Figure 1 — the revenue bridge: one gain, two larger losses.

SECTION 2

# The decision this supports

**The question**

Whose plan gets rewritten next quarter — retention, pricing, or merchandising?

An analysis is only worth reading to the extent it changes who does what on Monday. This one converts a single headline — revenue fell — into four named owners with a quantified problem, an expected recovery and a review date. If it does not change someone's plan next quarter, it has failed.

## What this report is, and what a dashboard is

| | This report | The dashboard that follows it |
|---|---|---|
| Job | explain a change and support a decision | monitor the KPIs the analysis identified |
| Question | “why, and what should we do?” | “are we OK?” |
| Cadence | one-off, on demand | monthly, always-on |
| Output | findings + recommendations, with owners | tiles, trends, exceptions |

The dashboard ships after the thinking and monitors what this report concluded. Its build specification, including the DAX measures and the targets below, is docs/07_dashboard_spec.md.

SECTION 3

# What the data is, and what it is not

Line-item grain: one row = one line item on one customer receipt — 195,224 lines across 55,431 invoices in the two quarters, plus a separate branch x SKU x day shelf-availability table that makes the supply event provable rather than plausible.

## Declared limits — limits on the conclusions, not footnotes

- 0.76% of revenue carries no customer identifier (KSh 0.70M of anonymous walk-ins). Every segment-level conclusion therefore reconciles to 99.24% of revenue, not 100%. The reconciliation is shown as its own line rather than hidden in an 'Other' bucket.
- The dataset is simulated from a documented mechanism (docs/00_ground_truth.md). That is deliberate: it lets the diagnosis be graded. See outputs/validation_report.md — 14/14 checks pass against the sealed answer key.
- No procurement, lead-time or supplier-performance data exists in this extract. The fifth why therefore stops at a data-capture recommendation instead of a guess.
- The annualised figure is a stated run-rate (KSh -20.83M, the quarter's change x 4), not a forecast. It assumes the Mombasa Nyali incident does not repeat and ignores Q4 seasonality.

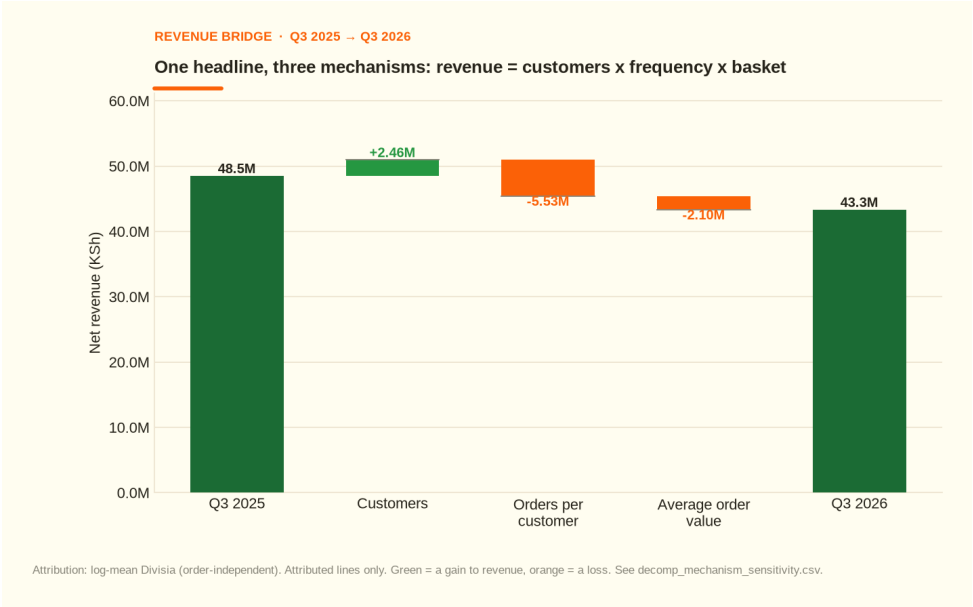
Method in one line: decompose by mechanism first (customers x frequency x basket), then by structure (store, category, segment, channel), then intersect the three, then ask why five times with a number attached to every answer. The attribution is log-mean Divisia, audited so the driver partition covers the decline exactly.

SECTION 4

# Descriptive — what happened

| Metric | Q3 2025 | Q3 2026 | Change |
|---|---|---|---|
| Net revenue | KSh 48.86M | KSh 43.65M | -10.7% |
| Gross profit | KSh 9.68M | KSh 8.04M | -16.9% |
| Gross margin | 19.95% | 18.57% | -1.38pp |
| Active customers | 5,265 | 5,555 | +5.5% |
| Orders | 28,618 | 26,764 | -6.5% |
| Orders per customer | 5.44 | 4.82 | -11.4% |
| Average order value | KSh 1,694 | KSh 1,619 | -4.5% |
| Lines per order | 3.59 | 3.40 | -5.2% |
| List price per unit | KSh 312 | KSh 326 | +4.3% |
| Discount depth (% of gross) | 4.80% | 7.82% | +3.02pp |

Every figure in this table is reproduced from outputs/kpis.json, which the pipeline regenerates. Nothing here is typed by hand.




*One customer gain against two larger losses. This is not one event.*

Figure 1 — the revenue bridge.

SECTION 5

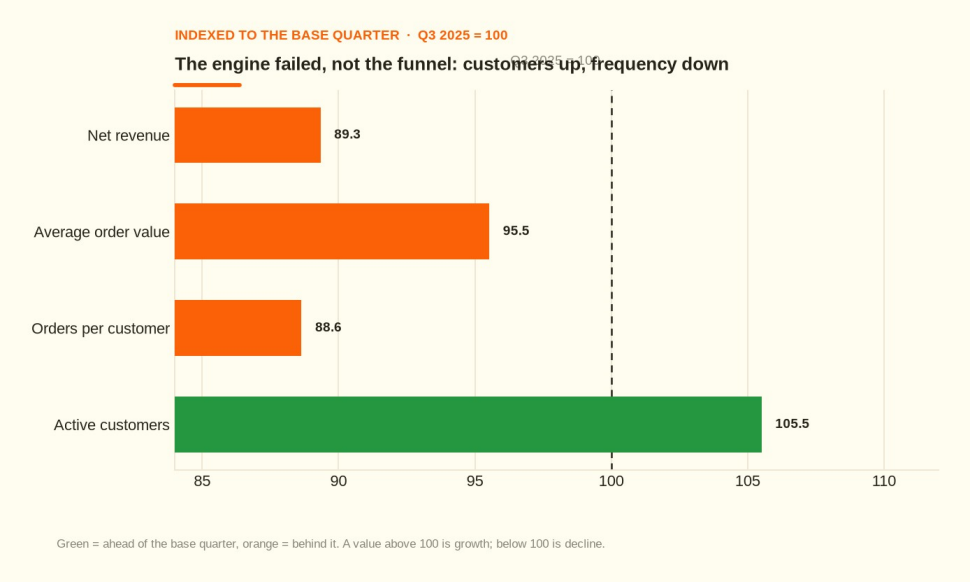
# Diagnostic, part 1 — the mechanism

**Revenue = Customers x Frequency x Average order value.**

| Driver | Effect on revenue | YoY move |
|---|---|---|
| Customers | KSh +2.46M | +5.5% |
| Orders per customer | KSh -5.53M | -11.4% |
| Average order value | KSh -2.10M | -4.5% |
| Total | KSh -5.21M | -10.7% |

Attribution is log-mean Divisia, so the ranking does not depend on the order in which the three factors are multiplied. The frequency effect is the largest under every ordering — that is the robust part of the result.

Sequential-order sensitivity band: customers KSh 2.26M to KSh 2.67M; frequency KSh -5.81M to KSh -5.26M; aov KSh -2.17M to KSh -2.03M.




*Customers grew. Orders per customer and basket value did not. Any report that stops at “fewer customers” is wrong.*

Figure 2 — the same three drivers indexed to the base quarter.

SECTION 6

# Diagnostic, part 2 — the structure, and who owns what

## By store

| Rank | Slice | Revenue change | YoY | Share of group decline |
|---|---|---|---|---|
| 1 | Mombasa Nyali | KSh -2.18M | -27.0% | 41.9% |
| 2 | Thika Road | KSh -1.50M | -18.7% | 28.9% |
| 3 | Kisumu Lakefront | KSh -0.69M | -11.5% | 13.2% |
| 4 | Town Centre | KSh -0.39M | -3.4% | 7.4% |
| 5 | Westlands | KSh -0.26M | -2.8% | 5.0% |
| 6 | Nakuru CBD | KSh -0.19M | -3.0% | 3.6% |

## By category

| Rank | Slice | Revenue change | YoY | Share of group decline |
|---|---|---|---|---|
| 1 | Groceries | KSh -1.60M | -13.1% | 30.7% |
| 2 | Beverages | KSh -1.08M | -20.3% | 20.7% |
| 3 | Household & Cleaning | KSh -0.82M | -11.3% | 15.7% |
| 4 | Personal Care | KSh -0.65M | -7.9% | 12.4% |
| 5 | Baby & Kids | KSh -0.46M | -6.7% | 8.8% |
| 6 | Dairy & Eggs | KSh -0.37M | -8.0% | 7.0% |
| 7 | Fresh Produce | KSh -0.15M | -4.7% | 2.9% |
| 8 | Bakery | KSh -0.09M | -6.9% | 1.7% |
| 9 | Unclassified | KSh -0.00M | -27.9% | 0.0% |

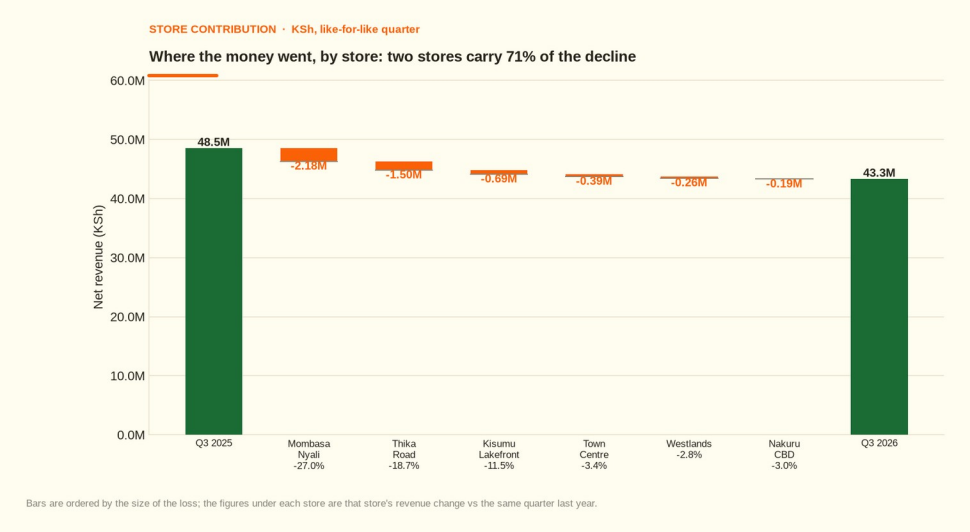
## By customer segment

| Rank | Slice | Revenue change | YoY | Share of group decline |
|---|---|---|---|---|
| 1 | Loyal | KSh -3.03M | -15.2% | 58.2% |
| 2 | Regular | KSh -2.52M | -14.0% | 48.4% |

| Rank | Slice | Revenue change | YoY | Share of group decline |
|---|---|---|---|---|
| 3 | At-Risk | KSh -0.85M | -14.2% | 16.3% |
| 4 | Hibernating | KSh -0.04M | -2.9% | 0.8% |
| 5 | New | KSh +1.28M | +40.7% | -24.5% |

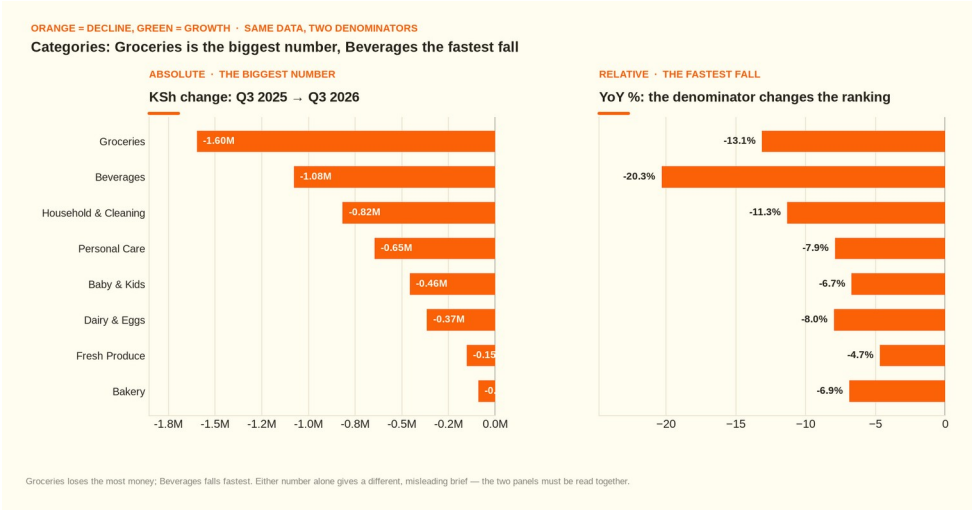
## By channel

| Rank | Slice | Revenue change | YoY | Share of group decline |
|---|---|---|---|---|
| 1 | In-store | KSh -4.54M | -10.8% | 87.1% |
| 2 | Online | KSh -0.67M | -9.9% | 12.9% |




*Two of six stores carry 71% of the decline.*

Figure 3 — where the money went, by store.

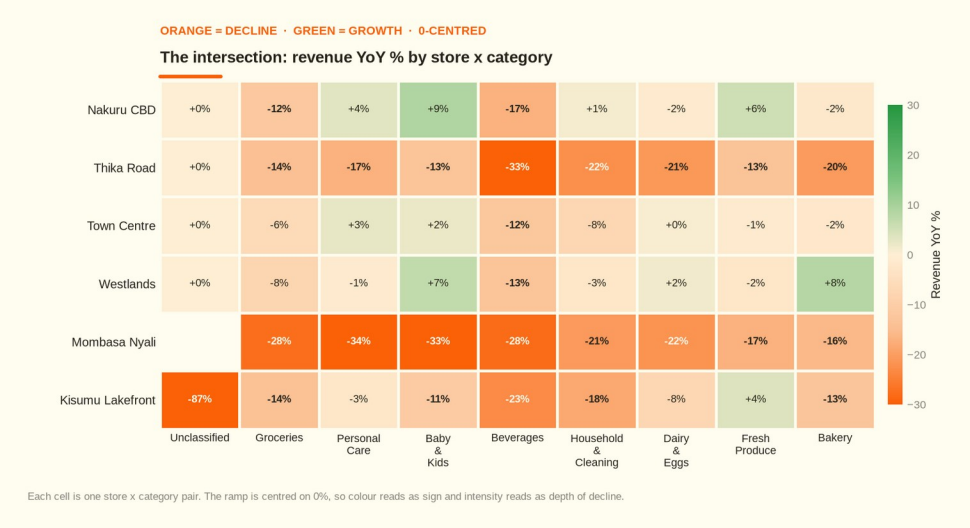



*Groceries is the biggest absolute loss, Beverages the fastest fall. Either alone is a misleading brief.*

Figure 4 — categories, in both KSh and %.

SECTION 7

# The intersection — which store x product x segment




*The decline is concentrated, not diffuse — which is why the fix can be targeted too.*

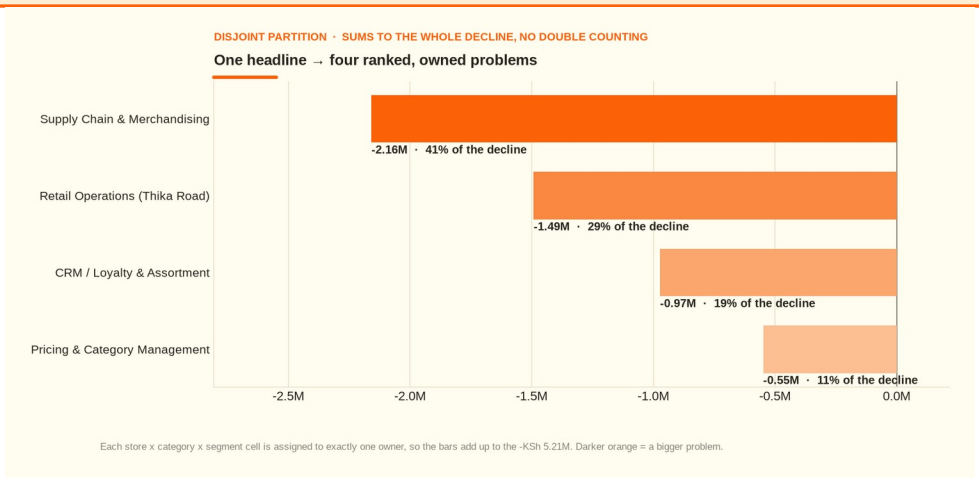
Figure 5 — revenue YoY % by store x category.

The worst 8 cells (store x category x segment) account for 23.5% of the total decline.

| # | Cell | Revenue change | YoY | % of group decline |
|---|---|---|---|---|
| 1 | Mombasa Nyali \| Groceries \| Regular | KSh -0.28M | -33.6% | 5.4% |
| 2 | Mombasa Nyali \| Groceries \| Loyal | KSh -0.23M | -29.6% | 4.4% |
| 3 | Mombasa Nyali \| Personal Care \| Regular | KSh -0.22M | -41.1% | 4.2% |
| 4 | Mombasa Nyali \| Baby & Kids \| Loyal | KSh -0.21M | -42.0% | 4.0% |
| 5 | Mombasa Nyali \| Personal Care \| Loyal | KSh -0.19M | -35.9% | 3.6% |
| 6 | Mombasa Nyali \| Baby & Kids \| Regular | KSh -0.17M | -34.8% | 3.3% |
| 7 | Thika Road \| Personal Care \| Loyal | KSh -0.17M | -30.9% | 3.3% |
| 8 | Mombasa Nyali \| Household & Cleaning \| Regular | KSh -0.17M | -37.0% | 3.2% |

### The counter-intuitive result

The single largest cell is Mombasa Nyali | Groceries | Regular — not the category that was out of stock. At Mombasa Nyali, every category fell together because the stock failure removed the reason for the trip and the whole basket shrank with it. Measuring only the affected category would have understated this event by roughly 4.8x.



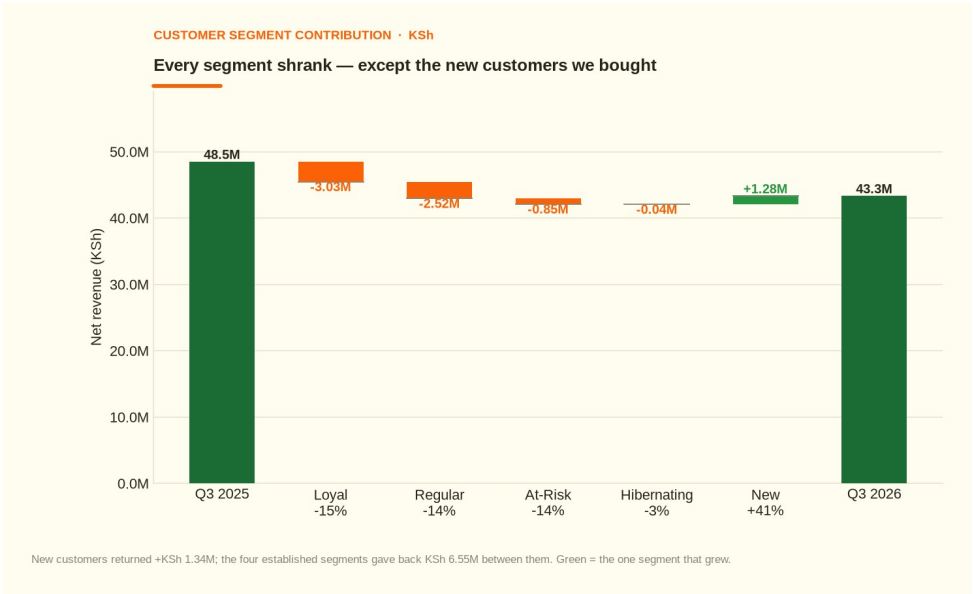

*Each cell is assigned to exactly one owner, so there is no double counting.*

Figure 6 — four ranked, owned problems (disjoint; sums to the whole decline).

SECTION 8

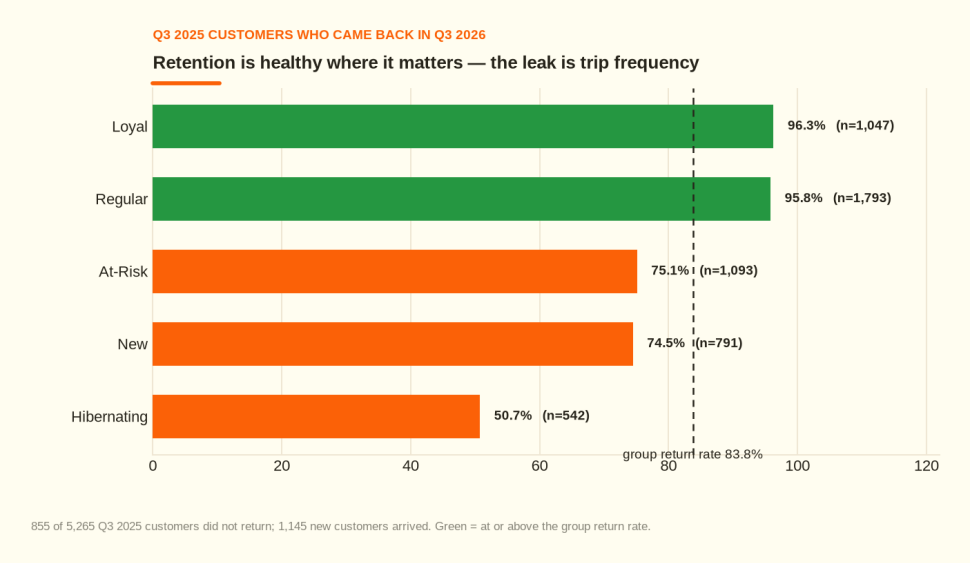
# Segments and retention

| Cohort metric | Value |
|---|---|
| Q3 2025 customers | 5,265 |
| Returned in Q3 2026 | 4,410 (83.8%) |
| Lapsed | 855 |
| New customers acquired | 1,145 |




*Buying growth is not a strategy while the existing base contracts.*

Figure 7 — every established segment shrank except the customers we bought.




*Retention is healthy where it matters. The leak is trip frequency, not churn.*

Figure 8 — return rate by base-quarter segment.

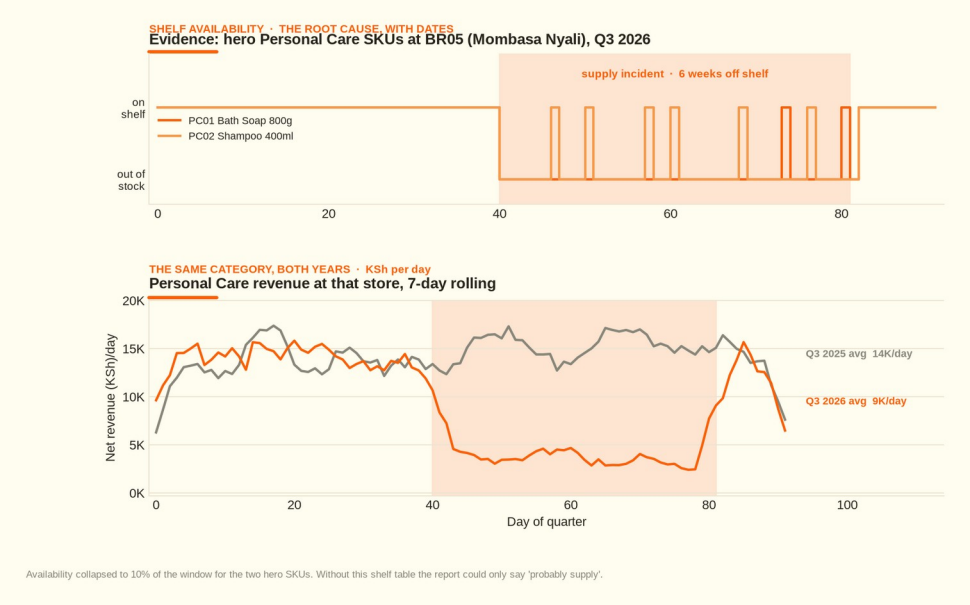
SECTION 9

# The supply event, with dates

| | Value |
|---|---|
| Store | Mombasa Nyali |
| Hero SKUs | Bath Soap 800g |
| Shelf availability in the incident window | 9.5% |
| Days out of stock | 76 |
| Total cost attributed | KSh 2.16M |
| Of which the affected category | KSh 0.45M |
| Of which basket spillover (other categories) | KSh 1.71M |

**Why this is evidence, not a story**

The shelf-availability table records, per branch, per SKU, per day, whether the item was on the shelf. That is what turns “probably a supply problem” into a dated, store-specific, owned finding.




*The incident has a start date, an end date, a store and a measurable cost.*

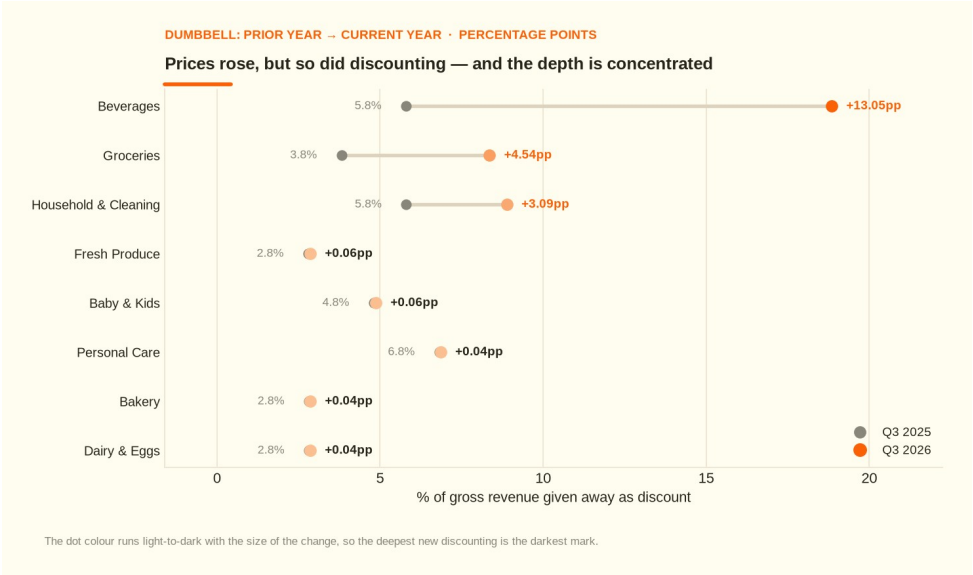
Figure 9 — shelf availability and the revenue trace that followed it.

SECTION 10

# Pricing, mix and margin

| | Q3 2025 | Q3 2026 | Change |
|---|---|---|---|
| Discount depth (% of gross) | 4.80% | 7.82% | +3.02pp |
| Gross margin | 19.95% | 18.57% | -1.38pp |
| Gross profit | KSh 9.68M | KSh 8.04M | -16.9% |

Roughly KSh 1.43M of extra discount was given away this quarter, concentrated in Beverages (+13.05pp). Gross profit fell faster than revenue, so margin — not just revenue — is the thing at risk.




*Prices rose, but so did discounting. The depth is concentrated.*

Figure 10 — discount depth, prior year vs current year, by category.

SECTION 11

# Facts, hypotheses, and the edge of what we know

A senior analysis is judged as much by what it refuses to claim as by what it concludes. Every statement below is tagged.

| Statement | Tier | Note |
|---|---|---|
| Revenue fell 10.7%; gross profit fell 16.9% | Fact | measured from the cleaned star schema |
| Two stores explain 71% of the decline | Fact | measured; contribution shares sum to the group change |
| Hero SKUs were 9.5% available for 76 days at Mombasa Nyali | Fact | from the branch x SKU x day availability table |
| Active customers rose +5.5% while frequency fell -11.4% | Fact | the mechanism identity, order-independent attribution |
| The stock-out *caused* the basket erosion at that store | Hypothesis | the two co-move exactly in time; no causal test was run |
| Single-sourcing or no safety stock caused the stock-out | Hypothesis | no procurement, supplier or lead-time data in this extract |
| Thika Road's decline is due to the road-works | Hypothesis | the frequency-only signature is consistent with an access problem; no other local factor is recorded |
| Kisumu Lakefront's -11.5% | Unexplained | flagged, deliberately not back-filled with a story |

| Statement | Tier | Note |
|---|---|---|
| 855 Q3 2025 customers did not return | Fact | but 83.8% did — churn is not the problem |

SECTION 12

# Predictive — the run-rate if nothing changes

**A run-rate, not a forecast**

If this quarter's frequency and basket levels persist, the annualised revenue impact is roughly KSh -20.83M (KSh 5.21M x 4). This assumes the Mombasa Nyali incident does not repeat — it should be fixed — and it ignores Q4 seasonality, which is typically stronger. Treat it as the size of the hole, not the shape of the year.

The honest reading: the quarter's damage is stable and repeatable in nature, so the cost of inaction is not zero. It is the run-rate above, minus whatever the actions in Section 13 recover.

SECTION 13

# Prescriptive — what to do

Every action carries the four things an action needs to be real: an owner, an expected impact, an evidence link and a review date. Impacts are deliberately conservative and priced in gross profit — they assume partial capture.

| # | Action | Owner | Why (evidence) | Expected impact | Review |
|---|---|---|---|---|---|
| 1 | Rebuild safety stock on the two hero Personal Care SKUs and qualify a second supplier | Supply Chain | Availability was 9.5% for 76 days; the store lost KSh 2.18M | KSh 0.28M – KSh 0.40M | 2026-11-04 (+30d) |
| 2 | Rebuild the Mombasa basket: substitutes on shelf, Personal Care adjacent to Groceries, a substitution script for staff | Merchandising | AOV fell -22.6% while the category alone explains only KSh 0.45M of the KSh 2.16M damage | KSh 0.03M | 2026-11-19 (+45d) |
| 3 | Thika Road access recovery: landlord and county engagement, extended hours, local marketing to restore trips | Retail Operations | Frequency fell -23.7% while basket value held at -0.3% — a pure traffic problem worth KSh 1.49M | KSh 0.17M | 2026-12-04 (+60d) |
| 4 | Cap beverage promotional depth back toward 4.8% | Pricing & Category Management | Beverage discount depth rose +13.05pp; category | KSh 0.27M | 2026-10-26 (+21d) |

| # | Action | Owner | Why (evidence) | Expected impact | Review |
|---|---|---|---|---|---|
| | of gross and require an approval gate above it | | revenue fell -20.3% | | |
| 5 | Convert the frequency leak: targeted win-back for the 855 lapsed customers and a second-purchase nudge for the 1,145 new ones | CRM / Loyalty | -11.4% trips per customer across all four established segments; residual pool worth KSh 0.97M | KSh 0.09M | 2027-01-03 (+90d) |
| 6 | Capture a customer identifier on every transaction; start logging supplier promised-vs-actual lead times | Data / POS & Supply Chain | 0.76% of revenue is currently unattributable and the root cause of the stock-out is unanswerable | No direct revenue; removes both blind spots | 2027-01-03 (next quarter) |

## Sequencing and dependencies

- Days 0-30: actions 1 and 4. Both are cheap and reversible, and both stop active bleeding. Action 4 is a policy change; action 1 is a purchase order.
- Days 30-60: actions 2 and 3. These need time to show up in traffic data.
- Days 60-90: action 5, which depends on knowing the base is stable first.
- Continuous: action 6. It is the only item that makes the *next* diagnosis cheaper — and this entire report exists because two data gaps had to be worked around.

SECTION 14

# How we will know whether it worked

Track the same five numbers this analysis used, on the same grain, every month. An analysis that never checks its own predictions never improves.

| Metric | Baseline | Target by Q1 FY2027 | Watched by |
|---|---|---|---|
| Orders per customer | 5.44 | back above 5.44 | CRM |
| Hero SKU shelf availability (Mombasa) | 9.5% | >99% | Supply Chain |
| Thika Road visits per customer | -23.7% YoY | >0% YoY | Retail Ops |
| Beverage discount depth | 7.82% of gross | <= 4.80% | Pricing |
| Gross margin | 18.57% | back above 19.95% | Finance |

APPENDIX A

# Method, provenance and reproducibility

The whole project is a pipeline, not a set of notebooks. Each step consumes the previous step's outputs, so the analysis is reproducible rather than narrated.

| Step | What it does |
|---|---|
| generate_data | simulate the star schema, with planted data-quality defects |
| clean_data | audit and repair, and write the data-quality report |
| run_sql | build the SQLite warehouse and run every sql/*.sql |
| decompose | mechanism, structure and the disjoint driver partition |
| five_whys | the evidential why-chain, generated from tables |
| write_report | the findings report and the recommendations register |
| make_figures | the ten decision charts, in the dashboard's visual language |
| write_reports | this report, the brief, the deck and the PDF |
| validate_against_truth | grade the diagnosis against the sealed answer key |

Reproduce everything with: python3 src/run_pipeline.py (dataset seed 20260914). Note on the interpreter: the vendored numpy is built for CPython 3.12, so run the pipeline with a 3.12 interpreter that can see pylibs/ — for example make PY=/usr/bin/python3.12.

The dataset is simulated from a documented mechanism (docs/00_ground_truth.md). That is deliberate: it allows the diagnosis to be graded. outputs/validation_report.md records 14/14 falsifiable checks passing.

APPENDIX B

# Index of charts

| Figure | Section | The sentence it proves |
|---|---|---|
| One headline, three mechanisms | Mechanism | The 10.7% fall is one customer gain against two larger losses — it is not one event. |
| Customers up, frequency down | Mechanism | Acquisition is working; repeat behaviour is not. Different problem, different owner. |
| Where the money went, by store | Structure | Two stores carry 71% of the decline. |
| The intersection: store x category | Intersection | The decline is concentrated, not diffuse — so the fix can be too. |
| Categories: KSh and % side by side | Structure | Groceries is the biggest absolute loss, Beverages the fastest fall. Either alone |

| Figure | Section | The sentence it proves |
|---|---|---|
| | | misleads. |
| Every segment shrank except the customers we bought | Segments | Buying growth is not a strategy while the existing base contracts. |
| Who came back | Segments | 83.8% returned — the leak is trips, not churn. |
| The supply failure, with dates | Evidence | 9.5% availability for 76 days at Mombasa Nyali. |
| Prices rose, discounting rose faster | Pricing | KSh 1.43M of extra discount in one quarter. |
| Four ranked, owned problems | Actions | Disjoint and audited: the bars sum to the whole decline with no double counting. |

APPENDIX C

# The evidential 5-Whys

Every “why” is answered with a number and a source, and the chain stops where the data stops rather than speculating.

**Why 1 — Why did group revenue fall?**

Net revenue fell KSh 5.17M (10.66%), from KSh 48.49M to KSh 43.32M. Volumes, not customers: active customers ROSE +5.51% while orders fell 6.48%.

Source: mechanism_by_period.csv · No — this is the question, not an answer

**Why 2 — Customers grew but orders fell. Why?**

Orders per customer fell -11.36% (5.44 -> 4.82 trips), and average order value fell -4.48% (KSh 1,694 -> KSh 1,619). Log-mean attribution puts KSh 2.46M of growth on new customers, KSh 5.53M against frequency and KSh 2.10M against basket value.

Source: mechanism_by_period.csv, decomp_mechanism_waterfall.csv · Partly — it names the mechanism (repeat visits, not acquisition)

**Why 3 — Where is the frequency and basket loss concentrated?**

Two of six stores explain 71.3% of the entire group decline: Mombasa Nyali (KSh -2.18M, -27.0%) and Thika Road (KSh -1.50M, -18.7%). Frequency fell -11.7% at Mombasa and -23.7% at Thika Road, versus -11.4% for the group.

Source: structure_by_branch.csv, structure_branch_mechanism.csv · Yes for the stores — but not yet for the cause

**Why 4 — Why did Mombasa Nyali lose so much more than the rest of the chain?**

Its two hero Personal Care SKUs were off the shelf for six weeks: 76 out-of-stock days out of 84 observations (9.5% shelf availability, first gap 2026-08-10). Personal Care revenue at that store fell -34.2% — but Mombasa's average order value also fell -22.6% (KSh 1,724 -> KSh 1,334) against a group AOV change of -4.5%. The missing items cost the whole basket, not just the category.

Source: fact_stockouts, stockout_incident_vs_rest.csv, structure_category_mechanism.csv · Yes — this is a supply-chain event with a date and a store

**Why 5 — Why were the SKUs unavailable for six weeks?**

The data shows WHAT happened (a 40-day gap on the worst single SKU, longest consecutive outage 33 days) but cannot show WHY. Replenishment, supplier and lead-time data are not in this dataset. Candidate causes — single-

sourced supply, no safety stock, a mis-set reorder point — are HYPOTHESES and must be labelled as such until procurement data is pulled.

Source: fact_stockouts (WHY is absent by design) · Stop here: the actionable unit is 'restore stock and dual-source', not 'explain the universe'